

Integrating Simple Rate Equations

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Abstract

I can't do calculus to save my life so I wrote out this document for myself so that I could remind myself how to integrate rate equations and use the integrated rate equations. Perhaps you will find it useful too.

I also wanted to explore using the $\text{\TeX}$ language to typeset the document. $\text{\TeX}$ is a great way to generate documents that have a beautiful professional appearance. It is used by mathematicians and physicists to create papers that contain complex math formulas. I used the $\text{\LaTeX}$ implementation of $\text{\TeX}$ and the $\text{\LyX}$ document processor to create the $\text{\LaTeX}$ code. $\text{\LyX}^1$ is a very useful program and enabled me to use $\text{\LaTeX}$ without needing to know the $\text{\TeX}$ markup language.

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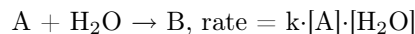
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¹ You can obtain $\text{\LyX}$ and the $\text{\LaTeX}$ system for most computer systems at <http://www.lyx.org>. See “*Tutorial #26: Introducing $\text{\LaTeX}$* ” for more details.

1 First Order Reactions

1.1 The Rate Equation

So for the simple case of a single reactant going to product we will consider a hydrolysis reaction.



If the water is the solvent and the concentration of A is a reasonable value, then the concentration of water will not change significantly during the reaction (a large number minus a small number is approximately equal to the original large number.) So the reaction can be considered to be...



We have incorporated the two constants, k and $[H_2O]$ (remember, we are saying $[H_2O]$ does not change) into a single observed rate constant we are calling k_{obs} . We know $[H_2O]$ because we have set up the reaction and so we can always obtain the value for k .

So let us define a rate equation to integrate. I will use k instead of k_{obs} for simplicity, but the k you see in the following equations is k_{obs} , and not the original k . I will also drop using the square brackets and use A and B instead of $[A]$ and $[B]$ for concentrations of reactants and products.

$$-\frac{dA}{dt} = k \cdot A \quad (1)$$

This is the rate of the reaction at any moment in time. But we started this reaction with a defined original concentration of A. We will call this initial concentration A_0 . We can say that x is the amount of A that has been converted to product(s) and so we state that...

$$A = A_0 - x \quad (2)$$

and so...

$$\frac{dx}{dt} = k \cdot (A_0 - x) \quad (3)$$

1.2 Integrating The Rate Equation

We can now rearrange the differential equation so that variables and their derivative are all collected together.

$$\frac{1}{A_0 - x} \cdot dx = k \cdot dt \quad (4)$$

Now we integrate both sides of the equation...

$$\int \frac{1}{A_0 - x} \cdot dx = \int k \cdot dt \quad (5)$$

to get...

$$-\ln(A_0 - x) = k \cdot t + c \quad (6)$$

where c is the constant of integration. But what is c? We can't just leave it as "c", we need to find its value. So we will evaluate the equation using the one data point that we know for sure: at $x = 0, t = 0$.

$$\begin{aligned} -\ln(A_0 - 0) &= k \cdot t + c \\ -\ln(A_0) &= c \end{aligned} \quad (7)$$

And so we can substitute the value for c into equation 6 to get...

$$-\ln(A_0 - x) = k \cdot t - \ln(A_0) \quad (8)$$

which rearranges to give...

$$\ln\left(\frac{A_0 - x}{A_0}\right) = -k \cdot t \quad (9)$$

We can make the equation to plot x vs. t by performing the exponential function on both sides of the equation 9.

$$A_0 - x = A_0 \cdot e^{-k \cdot t} \quad (10)$$

Recall that $[A_0] - x = [A]$ and that $x = [B]$. So equation 10 can give the equations for $[A]$ and $[B]$ in terms of $[A_0]$, k_{obs} and time.

$$A = A_0 \cdot e^{-k \cdot t} \quad (11)$$

and

$$\begin{aligned} B &= A_0 - A_0 \cdot e^{-k \cdot t} \\ B &= A_0 \cdot (1 - e^{-k \cdot t}) \end{aligned} \quad (12)$$

1.3 Using The Integrated Rate Equation

The **half life** of the reaction is defined as the time at which the reaction is 50% complete (as defined in this case by disappearance of A).

At $t_{\frac{1}{2}}$ we know that $x = \frac{1}{2} [A_0]$ so we can say that...

$$\ln\left(\frac{A_0 - \frac{1}{2} \cdot A_0}{A_0}\right) = -k \cdot t$$

which solves to give

$$t = \frac{\ln(2)}{k}$$

To create a **linear form of the integrated equation** for line fits we can use equation 9 directly. Recall that $[A_0] - x = [A_0]$.

We can use the measured concentration of A to plot $\ln(A)$ vs. t to get a straight line. This line can be fit to...

$$\ln(A) = \ln(A_0) - k \cdot t \quad (13)$$

Using this equation we can determine $\ln(A_0)$ and k_{obs} directly from the graph.

1.4 Some Thoughts On The Integration

Where did dA go when we substituted A with $A_0 - x$ in equation 3?

Recall that the derivative of a constant is zero. So we have the following set of statements...

$$-\frac{dA}{dt} = -\frac{d(A_0 - x)}{dt} = -\frac{dA_0 - dx}{dt} = -\frac{0 - dx}{dt} = \frac{-dx}{dt} = \frac{dx}{dt}$$

Shouldn't there have been a "constant of integration" on both sides of the equation in equation 6?

Yes, we got...

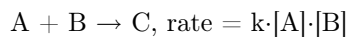
$$\begin{aligned} -\ln(A_0 - x) + c_1 &= k \cdot t + c_2 \\ -\ln(A_0 - x) &= k \cdot t + (c_2 - c_1) \end{aligned}$$

So you see that all the integration constants can be collected on one side of the equation or the other.

2 Second Order Reactions: $A + B$

2.1 The Rate Equation

For this we will consider a reaction where two reactants combine to form a product or products.



We can say that...

$$-\frac{dA}{dt} = -\frac{dB}{dt} = \frac{dC}{dt} = k \cdot A \cdot B \quad (14)$$

2.2 Integrating The Rate Equation

If A_0 and B_0 are defined as the initial concentrations of the reactants and x is the amount of each consumed in the reaction then we can establish our differential equation as follows...

$$\frac{dx}{dt} = k \cdot (A_0 - x)(B_0 - x) \quad (15)$$

Collecting the terms gives and integrating the equation gives us...

$$\int \frac{dx}{(A_0 - x)(B_0 - x)} = \int k \cdot dt$$

which results in...

$$\frac{\ln(A_0 - x) - \ln(B_0 - x)}{A_0 - B_0} = k \cdot t + c \quad (16)$$

By evaluating the equation at $t=0$, $x=0$ we can find the value of c .

$$c = \frac{\ln(A_0) - \ln(B_0)}{A_0 - B_0}$$

substituting this into equation 16 gives...

$$\frac{\ln(A_0 - x) - \ln(B_0 - x)}{A_0 - B_0} = k \cdot t + \frac{\ln(A_0) - \ln(B_0)}{A_0 - B_0} \quad (17)$$

which simplifies to give

$$\ln \frac{(A_0 - x) \cdot B_0}{(B_0 - x) \cdot A_0} = (A_0 - B_0) \cdot k \cdot t \quad (18)$$

We can obtain an equation that relates concentrations of reactants to time by applying the exponential function to both sides of the equation.

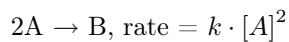
$$\frac{A_0 - x}{B_0 - x} = \frac{A_0}{B_0} \cdot e^{(A_0 - B_0) \cdot k \cdot t} \quad (19)$$

Note that $A_0 - x = A$ and $B_0 - x = B$ and that $x = C$.

3 Second Order Reactions: A + A

3.1 The Rate Equation

Let us consider a reaction like a dimerization where we have the following scheme.



We can say that

$$-\frac{dA}{dt} = k \cdot A^2$$

If we consider x to be the amount of A consumed in the reaction then we can state that...

$$-\frac{dx}{dt} = k \cdot (A_0 - x)^2 \quad (20)$$

3.2 Integrating The Equation

After collecting the terms and integrating we get...

$$\int \frac{dx}{(A_0 - x)^2} = \int k \cdot t$$

which results in...

$$\frac{1}{A_0 - x} = k \cdot t + c \quad (21)$$

and we evaluate the equation at t = 0, x = 0 and find c to be...

$$c = \frac{1}{A_0}$$

and so we substitute into equation 21 to get...

$$\frac{1}{A_0 - x} = k \cdot t + \frac{1}{A_0}$$

so

$$\frac{1}{A_0 - x} - \frac{1}{A_0} = k \cdot t \quad (22)$$

Recall that $A_0 - x = A$ so we could also state that...

$$\frac{1}{A} - \frac{1}{A_0} = k \cdot t \quad (23)$$

Linear plot: We can plot 1/A vs. t and fit it to the following equation. The y-intercept is 1/A₀ and the slope is k.

$$\frac{1}{A} = \frac{1}{A_0} + k \cdot t$$

Plotting Concentration of A vs. time: We can solve equation 23 for A and get...

$$A = \frac{A_0}{1 + A_0 \cdot k \cdot t} \quad (24)$$

4 Simple Parallel Reaction

We are fast leaving behind the trivial subjects of integrating rate equations. For example, rather than integrate the rate equation for a consecutive reaction, e.g. $A \rightarrow B \rightarrow C$, we prefer to use the steady state or equilibrium approximations to evaluate the rate law. This consecutive reaction can be solved using integration, its just complicated.

4.1 The Rate Equation

A bit simpler than the consecutive case is the case of parallel reactions where a reactant can be the source of two different products.



We can say that...

$$-\frac{dA}{dt} = k_1 \cdot A + k_2 \cdot A = (k_1 + k_2) \cdot A \quad (25)$$

4.2 Integrating The Equation

We can define x as the amount of A consumed in the reaction. So we say that...

$$\frac{dx}{dt} = (k_1 + k_2) \cdot (A_0 - x) \quad (26)$$

After collecting the term, integrating, solving for c, and simplifying we basically get the exact same equation as in the case of the first order reaction.

$$A = A_0 \cdot e^{-(k_1+k_2) \cdot t} \quad (27)$$

compare equation 26 to equation 11.

Now that we have an equation for A at any given time, we can evaluate a differential equation for B and C. The rate of formation of B and C depend on the [A] term. We can say that...

$$\frac{dB}{dt} = k_1 \cdot A \quad \text{and} \quad \frac{dC}{dt} = k_2 \cdot A \quad (28)$$

Let us begin with integrating an equation for [B]. We can substitute equation 27 into 28.

$$\frac{dB}{dt} = k_1 \cdot A_0 \cdot e^{-(k_1+k_2) \cdot t}$$

We collect the terms and integrate...

$$\int dB = k_1 \cdot A_0 \int e^{-(k_1+k_2) \cdot t} \cdot dt$$

we get...

$$B = -\frac{k_1}{k_1 + k_2} A_0 \cdot e^{-(k_1+k_2) \cdot t} + c$$

we evaluate c at t = 0, B = 0 and the integrated equation is obtained.

$$B = -\frac{k_1}{k_1 + k_2} A_0 \cdot e^{-(k_1+k_2) \cdot t} + \frac{k_1}{k_1 + k_2} A_0$$

which simplifies to...

$$B = -\frac{k_1}{k_1 + k_2} A_0 \cdot \left(1 - e^{-(k_1+k_2) \cdot t}\right) \quad (29)$$

Similarly, the integrated rate law for [C] is...

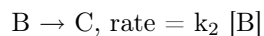
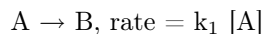
$$C = -\frac{k_2}{k_1 + k_2} A_0 \cdot \left(1 - e^{-(k_1+k_2) \cdot t}\right) \quad (30)$$

5 Simple Consecutive Reaction

5.1 The Rate Equation

Did I just say that integrating a rate law for consecutive reaction is difficult? It's difficult for me, but not for Maple. Maple is a math program available on any campus computer.

A consecutive reaction could be defined as...



We can say that...

$$-\frac{dA}{dt} = k_1 \cdot A \quad \text{and} \quad \frac{dB}{dt} = k_1 \cdot A - k_2 \cdot B$$

5.2 Integrating The Rate Equation Using Maple

The disappearance of A is a simple first order reaction and the rate law is given in equation 11. The concentration of B can be found by substituting the rate law for [A] into the differential equation for [B].

$$\frac{dB}{dt} = k_1 \cdot A_0 \cdot e^{-k_1 \cdot t} - k_2 \cdot B \quad (31)$$

Collecting the terms and integrating gives — wait a minute — this is hard! We can't easily separate all the B terms and the t terms to opposite sides the equation. We know the differential of [B] *w.r.t.* t (it is equation 31) so we need to solve for [B] such that the differential of [B] *w.r.t.* t is the expression defined in equation 31. I have long since completely forgotten my differential equations course. Fortunately, Maple hasn't.

We can solve the differential equation using the `dsolve` command in Maple. In fact, all the differential equations stated in this document could have been solved this way rather than collecting the terms, integrating and evaluating the constant of integration (which is the easiest way to solve a simple differential equation).

The `dsolve` command requires the differential equation and any known points in the function being evaluated. In this case we will use the B = 0 when t = 0 point. In Figure 1 on the following page we see a screen shot of Maple in action. See how easy it was to determine the function, B. The “%” symbol is a placeholder for the previous result. It feeds the previous result into that spot in the command. Learning how to use Maple is a very useful thing. I only know a handful of commands and it is a great help to me.

So we solved the equation below using the `dsolve` command...

$$\frac{d}{dt} B(t) = k_1 \cdot A_0 \cdot e^{-k_1 \cdot t} - k_2 \cdot B(t) \quad (32)$$

Note that I rewrote equation 31 a little bit differently to match up with the way it was presented in Maple. The concentration of B is a function of t so we wrote it as B(t).

The solution that Maple gave us was...

$$B(t) = -\frac{k_1 \cdot A_0 \cdot (e^{-k_1 \cdot t} - e^{-k_2 \cdot t})}{k_1 \cdot \ln(e) - k_2} \quad (33)$$

which simplifies to (remember that $\ln(e) = 1$, I have no idea why Maple left that in there)...

$$B(t) = -\frac{k_1}{k_1 - k_2} \cdot A_0 \cdot (e^{-k_1 \cdot t} - e^{-k_2 \cdot t}) \quad (34)$$

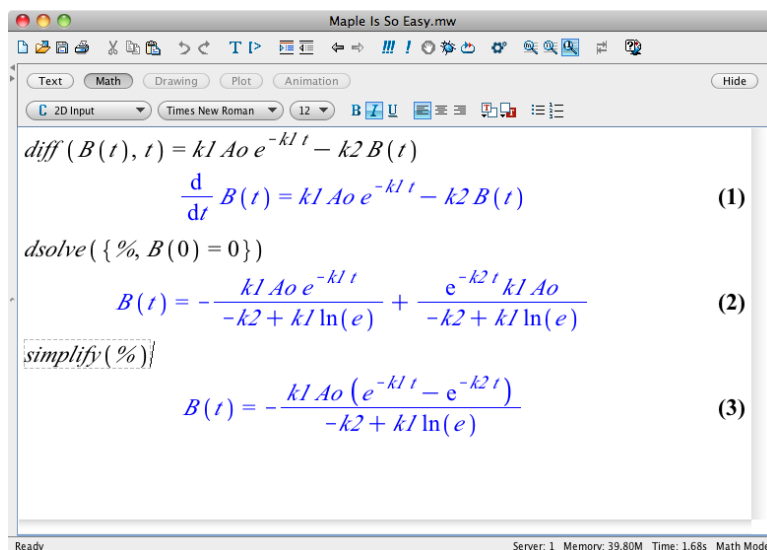


Fig. 1: Screen shot of Maple

We could also solve the differential equation for C using the `dsolve` command in Maple but fortunately we also know that $A_0 = A + B + C$ (matter cannot be created or destroyed in chemistry — usually). So...

$$C = A_0 - A - B \quad (35)$$

We substitute in equations 34 and 11 on page 3 to get...

$$C = A_0 - A_0 \cdot e^{-k_1 \cdot t} - \frac{k_1}{k_1 - k_2} \cdot A_0 \cdot (e^{-k_1 \cdot t} - e^{-k_2 \cdot t})$$

which simplifies to...

$$C = A_0 \cdot \left(1 - e^{-k_1 \cdot t} - \frac{k_1}{k_1 - k_2} \cdot A_0 \cdot (e^{-k_1 \cdot t} - e^{-k_2 \cdot t}) \right) \quad (36)$$

6 Maple Notes

Creating A Derivative: We defined the differential equation using the Maple `diff` command. So maple says that...

$$\text{diff}(B(x), x) = \frac{d}{dx} B(x)$$

You can't define a differential equation by entering d/dx as a fraction. The differential must be created using the `diff` command.